

DRYDEN, ON, Canada

WMO: 715270

Lat: 49.830N

Lon: 92.740W

Elev: 1354

StdP: 13.99

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-19.6	-31.2	1.0	-23.5	-26.6	1.3	-19.1	21.0	11.3	19.0	10.7	6.7	250	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	84.0	67.7	80.9	65.7	78.2	64.2	71.1	79.2	69.1	76.6	67.2	74.5	11.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.4	109.8	75.2	66.5	102.7	73.3	64.6	95.8	70.9	35.9	79.5	34.1	76.6	32.5	74.5	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	18.6	16.6	DB	-30.7	89.7	4.3	3.1	-33.8	91.9	-36.3	93.8	-38.7	95.5	-41.8	97.8
			WB	-31.0	74.6	4.2	2.0	-34.0	76.0	-36.5	77.1	-38.8	78.2	-41.9	79.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.3	4.2	7.5	22.4	37.0	50.6	61.4	66.9	64.5	55.6	41.0	25.7	9.3
	DBStd	24.45	14.05	12.88	13.66	10.45	9.45	6.94	5.96	6.30	8.72	9.06	11.58	13.33
	HDD50	6327	1421	1191	857	403	111	5	0	1	41	307	728	1262
	HDD65	10344	1886	1611	1320	841	455	147	49	86	301	743	1178	1727
	CDD50	1702	0	0	2	12	128	346	524	450	209	30	1	0
	CDD65	244	0	0	0	0	8	39	107	71	18	1	0	0
	CDH74	1769	0	0	0	2	81	310	779	480	114	3	0	0
	CDH80	361	0	0	0	0	11	52	179	102	17	0	0	0
	WSAvg	8.4	7.9	8.1	8.5	8.9	9.1	8.3	7.9	7.9	8.6	8.6	8.6	7.9
	PrecAvg	28.0	1.2	0.9	1.4	1.7	2.8	4.3	4.2	3.8	3.3	2.4	1.5	1.2
(15) Precipitation	PrecMax	35.4	3.9	2.5	3.5	4.6	6.3	8.5	7.9	7.7	6.9	5.4	4.1	2.4
	PrecMin	22.4	0.2	0.2	0.4	0.0	0.6	1.7	2.0	1.9	0.8	0.6	0.3	0.6
	PrecStd	4.1	0.8	0.4	0.7	1.2	1.5	1.7	1.4	1.5	1.5	1.3	0.7	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.8	41.1	60.2	69.0	81.2	85.8	89.1	87.3	82.6	71.5	56.3	37.7
		MCWB	34.5	37.7	49.3	52.3	60.9	69.5	70.8	70.6	67.5	58.1	49.6	34.8
	2%	DB	32.4	35.4	51.0	62.8	75.7	80.7	84.3	82.7	77.2	64.2	49.5	33.0
		MCWB	31.5	32.7	42.9	48.1	57.7	63.5	67.6	68.2	65.1	54.3	44.3	31.4
	5%	DB	28.5	31.7	44.5	58.1	71.3	77.5	81.5	79.1	73.0	59.5	44.6	31.0
		MCWB	27.4	29.3	37.9	45.3	55.6	62.3	66.5	65.9	62.8	51.8	41.0	29.9
	10%	DB	24.4	26.8	39.6	53.5	66.6	74.2	78.5	76.0	69.0	54.7	40.8	28.3
		MCWB	23.3	24.7	35.0	43.0	53.9	60.9	64.8	64.2	60.9	49.4	38.3	27.3
	0.4%	WB	34.7	38.1	53.4	54.5	66.2	72.3	73.8	73.8	70.3	60.6	51.0	35.4
		MCDB	36.1	40.9	57.1	63.0	74.9	81.1	83.6	82.7	78.4	66.4	55.0	36.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.7	33.1	44.1	50.1	62.2	68.3	71.4	70.5	67.5	56.1	45.3	32.1
		MCDB	32.6	35.0	50.5	59.0	69.0	76.1	79.9	78.1	73.6	63.2	48.8	32.9
	5%	WB	27.6	29.2	38.9	46.9	58.5	65.6	69.3	68.5	64.8	52.9	41.6	29.9
		MCDB	28.6	31.4	43.5	56.1	66.9	72.2	76.9	76.1	71.2	58.2	44.2	30.9
	10%	WB	23.3	24.6	35.3	44.0	55.7	63.0	67.4	66.4	62.1	49.5	37.9	27.2
		MCDB	24.3	26.4	39.1	52.9	64.2	70.4	74.8	73.5	67.3	53.6	40.8	28.3
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	17.9	18.8	18.8	19.1	17.9	17.8	17.4	15.9	12.4	11.2	13.3
		MCDBR	14.4	16.9	20.5	25.3	24.9	22.5	21.1	20.9	20.6	20.0	15.1	12.2
		MCWBR	13.4	14.0	13.5	14.8	11.5	9.5	8.8	9.2	10.6	12.1	10.7	11.1
	5% WB	MCDBR	13.9	14.7	19.2	22.5	20.2	17.8	18.0	18.0	18.2	18.0	14.0	11.3
		MCWBR	13.1	12.2	13.4	14.5	11.5	10.5	9.7	9.8	10.9	12.1	10.6	10.5
(40) Clear-Sky Solar Irradiance	taub		0.249	0.257	0.292	0.326	0.347	0.364	0.383	0.362	0.348	0.298	0.278	0.254
	taud		2.326	2.313	2.341	2.419	2.446	2.428	2.392	2.470	2.485	2.602	2.491	2.341
	Ebn at Noon		258	285	292	292	289	284	276	277	270	270	248	238
	Edh at Noon		21	27	32	33	34	35	36	31	28	21	18	18
(44) All-Sky Solar Radiation	RadAvg		374	758	1161	1564	1689	1750	1813	1521	1097	630	334	267
	RadStd		44	70	94	115	109	142	132	128	102	83	38	33

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-1.78	-1.66	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page